
Department of Geography, The University of Hong Kong

GEOG7305

Programming for Geospatial Data

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2nd term 2025 – 2026, Bo HUANG



Course Objectives

01

Understand the principles of GIS design using **Structured System Analysis and Design Methods**

02

Gain familiarity with the methods for customizing GIS

03

Develop competence in implementing a GIS-based application system



Topics to be Covered

Overview

- Introduction to GIS

Database Design

Programming

- Basics of Python
- Python for Geoprocessing

GIS Customization

- Dataset and layer management
- Attribute data management
- Raster data operation
- Spatial analysis
- Suitability analysis



Lab Exercise

Lab 01 Data Flow Diagram

- Report due on February 13

Lab 02 ER-Diagram

- Report due on February 27

Lab 03 Introduction to python

- Report due on March 6

Lab 04 Model builder

- Report due on March 20

Lab 05 Python for Geoprocessing

- Report due on April 10

Lab 06 Selection, Query, Data Management

- Report due on April 17



Important Schedules

3rd Week **February 6**
(Form Group)

6th Week **March 6**
(Detailed Proposal)

8th Week **March 20**
(Mid-term Pre)

13th Week **April 24**
(Final Pre)

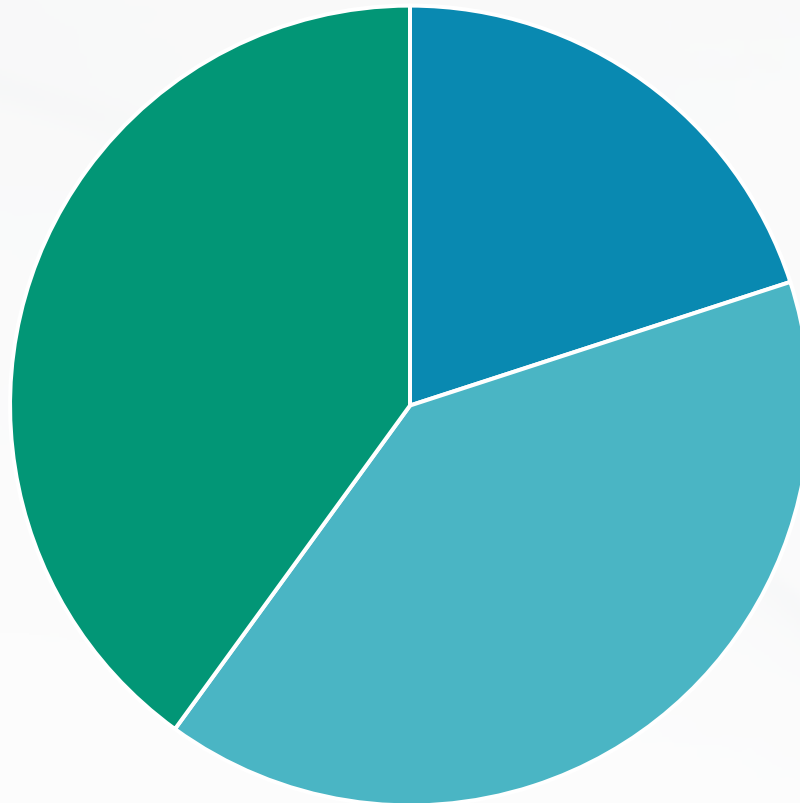
16th week **May 15** (Final Examination)

All project deliverables due on or before May 8



Grading Scheme

Grading Scheme



■ Lab Report ■ Group Project ■ Final Exam ■

01. Lab Report (20%)

02. Group Project (40%)

03. Final Examination (40%)



Class Discipline



Be punctual



Deactivate mobile phone



Do not talk before asked to



It is very important that you freely express your opinions, concerns, and comments regarding the lectures and lab sessions. Without them, it is unlikely that your concerns can be addressed.



Reference Books

- ❑ Paul A. Zandbergen, 2020. Python Scripting for ArcGIS Pro, ESRI Press.

Online materials :

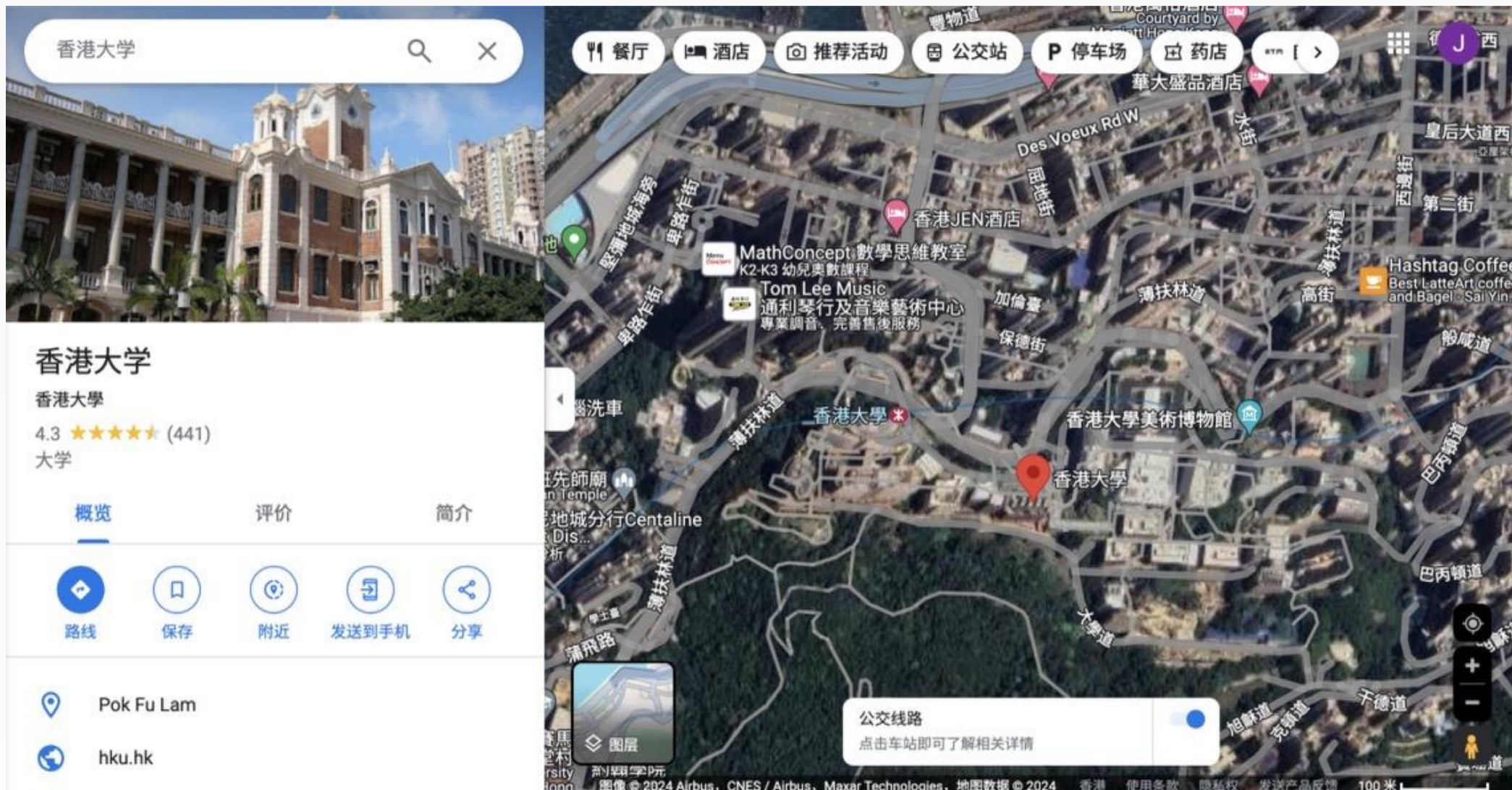
- ❑ Python in ArcGIS Pro: <https://pro.arcgis.com/en/pro-app/latest/arcpy/get-started/installing-python-for-arcgis-pro.htm>

01

GIS Application System Design

Background

GIS System





GIS System



What is Information System?

- A System to organize **INFORMATION**.





GIS System



What is Information System?

- A System to organize **INFORMATION**.

The screenshot shows the HKU FIND@HKUL search results page for the query 'GIS'. The page features a navigation bar with links to 'NEW SEARCH', 'HKU LIBRARIES', 'ELECTRONIC RESOURCES', and 'READINGLIST @HKUL'. A search bar at the top contains the text 'GIS'. Below the search bar, a yellow banner prompts 'HKU Patrons, please Sign in' with a 'DISMISS' button. The main content area displays search results for 'GIS', including a book by Nick Bearman published in 2020 by Bloomsbury Publishing. The book description states: 'This book provides a non-technical overview of the science and tools behind geographic information systems and geographic information science for researchers, students and academics who do not have a GIS...'. A 'Full text available' link is provided. On the right side, there are filters for 'Refine results', 'Sort by Relevance', 'Show Only' (with options for Full Text Online, Peer-reviewed Journals, Open Access, and Available in the Library), and 'Resource Type' (with options for Articles and Book Chapters).



GIS

- **Geographical Information System (GIS)**

- **I**

- Decision-making

- **GI**

- 80% of all data can be given spatial co-ordinates
 - “***Spatial thinking** as a universal mode of thinking Using the properties of space structuring problems, finding answer, expressing solutions*” (US National Research Council, Learning to Think Spatially, 2006)



GIS

- **IS**

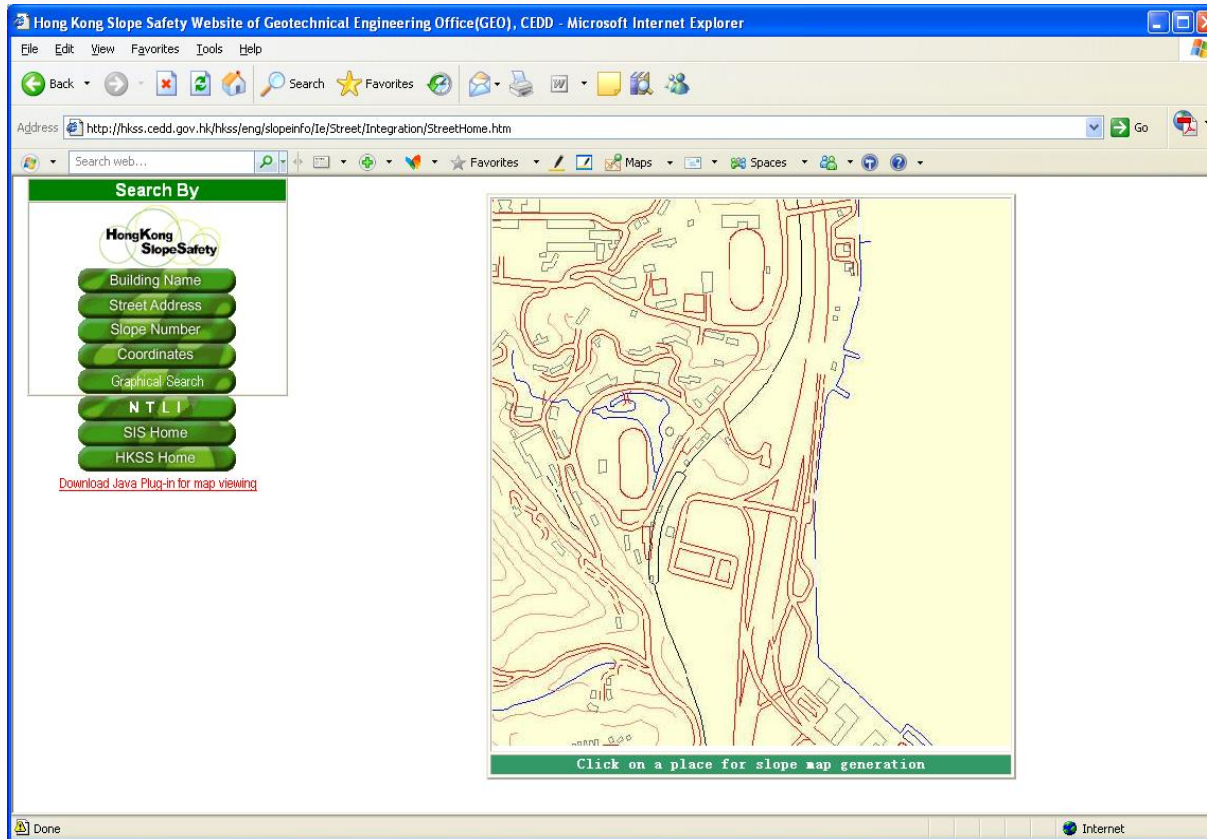
- The term information system (IS) sometimes refers to a system of persons, data records and activities that process the data and information in an organization, and it includes the organization's manual and automated processes (Wikipedia).

- **GIS**

- IS that integrate, store, edit, analyze, share, and display georeferenced information.

• Examples

- Slope Catalogue Database



- IS:
 - Data: Name, owner, location, maintenance records etc.
 - Function: Search by date last maintenance, region Etc.
- GIS:
 - Data: Geographic location, shape of the slopes
 - Functions:
 - Nearby slopes of a given location
 - Distribution of slopes maintained in last 5 years
 - Spatial patterns of dangerous slopes under given rainfall.

- Geographic information is much more complex and huge than ordinary textual information:
 - Textual information (I):
 - Types: Date, character, number, real number
 - Operators: +, -, x, /, IF-THEN, <, > etc.
 - Geographic information (GI):
 - Types: Point, line, polygon, network, TIN, raster etc.
 - Operations: distance, adjacent, buffer, shortest path, overlay, co-ordinate transformation, modelling etc.

- **GIS**

- IS that integrates, stores, edits, analyses, shares, and displays georeferenced information.

Technical Aspects

- GI Data Model
- GI Database
- GI Analysis
- GI Visualization

Technical – Human Aspects

- GI System Applications
- GI System Design and Analysis
- GI System Project management

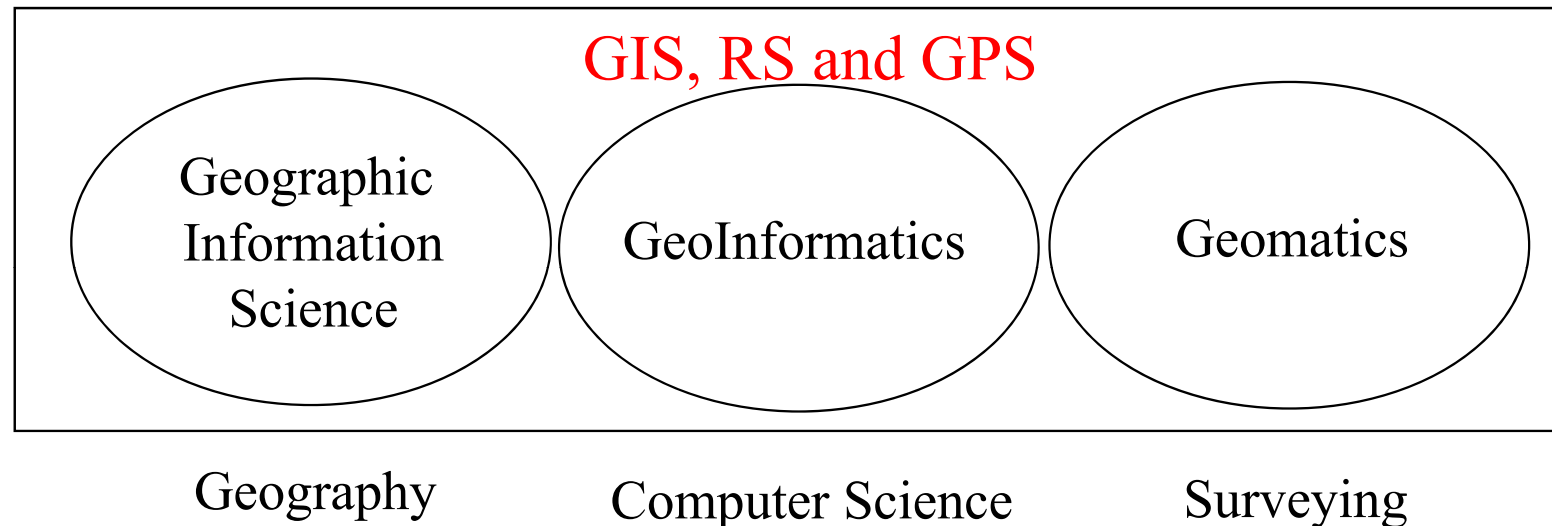


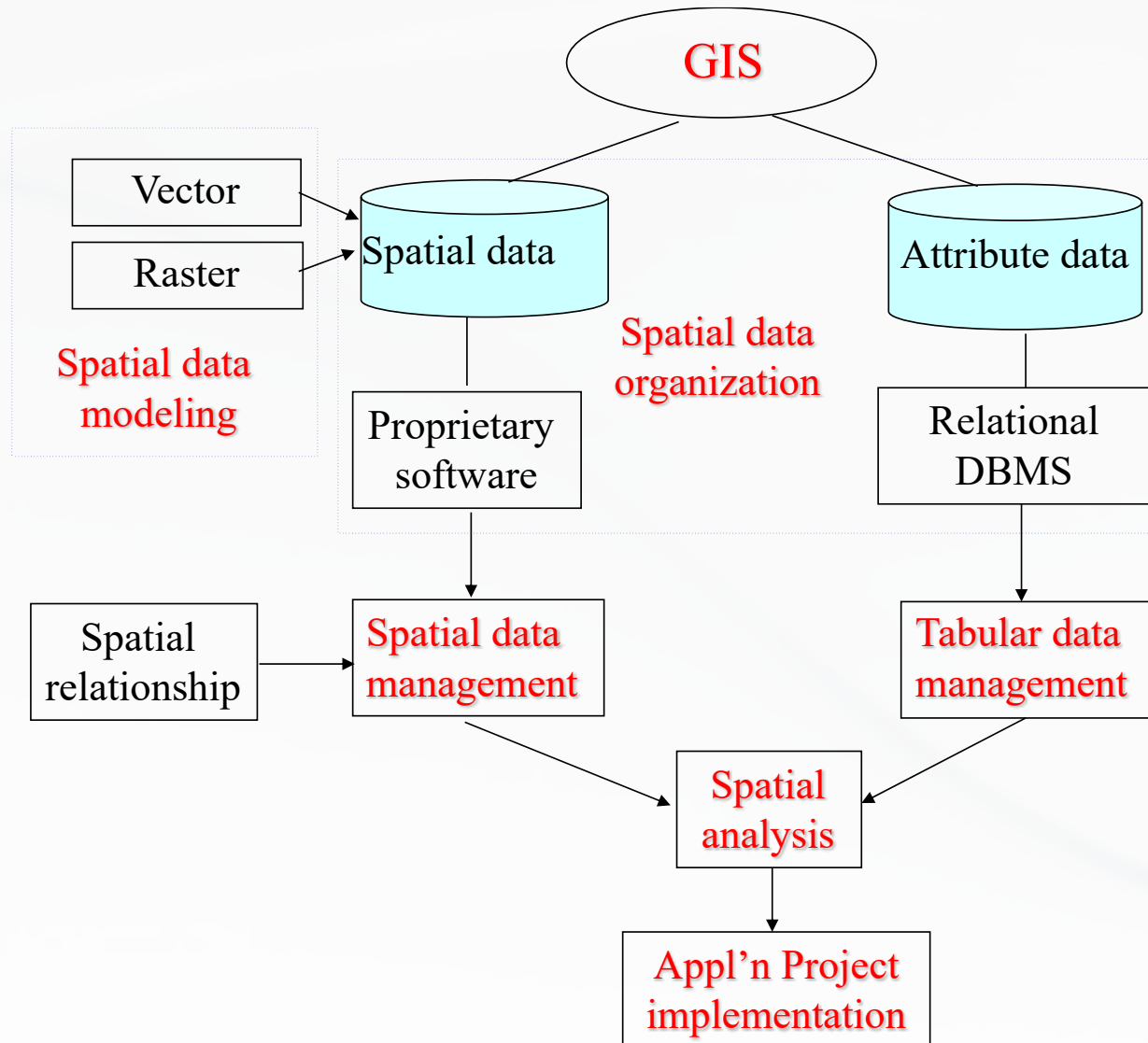
GIS

What is GIS?

G I S is a computer-based system that is used for **geospatial data** input, storage, management, analysis and presentation.

Multi-disciplinary Nature





Pillars of GIS

1. Architecture
2. Data models (Vector & Raster)
3. Queries (Spatial & Attribute & Mixed)
4. Analysis (Overlay & Buffer & Network & Terrain)

What can GIS do?

Spatial Data Input

Data entry
Data editing
Geometric
transformation
Projection

Data Management

Data
modification
Database
management

Data Display

Use of maps,
charts and
tables

Data Exploration

Attribute data
query
Spatial data
query
Geographic
visualization

Data Analysis

Geometric
analysis
Terrain
analysis
Network
analysis

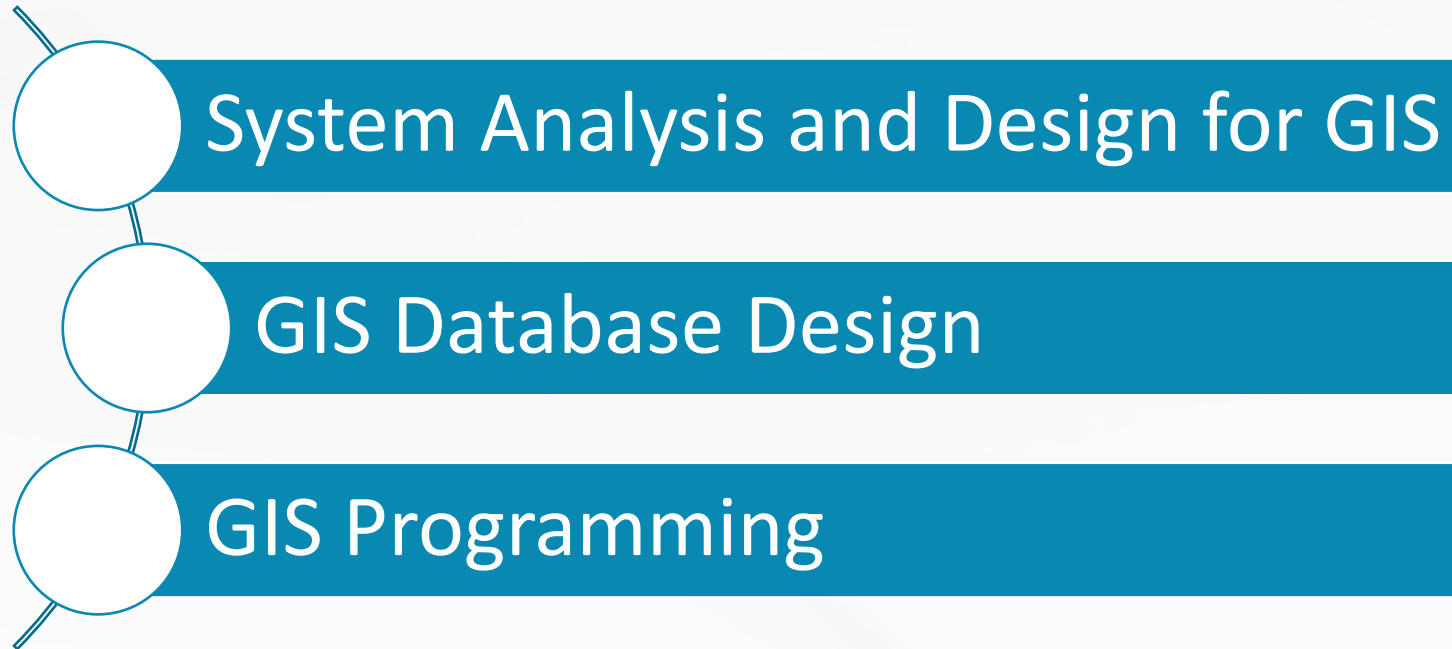


GIS Applications

- Mapping....
- Urban Planning.....
- Transportation Planning....
- Disaster Risk and Management....
- Agricultural Value Chain....
- Natural Resource Management.....
- Surveying....
- Public Health....



GIS Design and Implementation



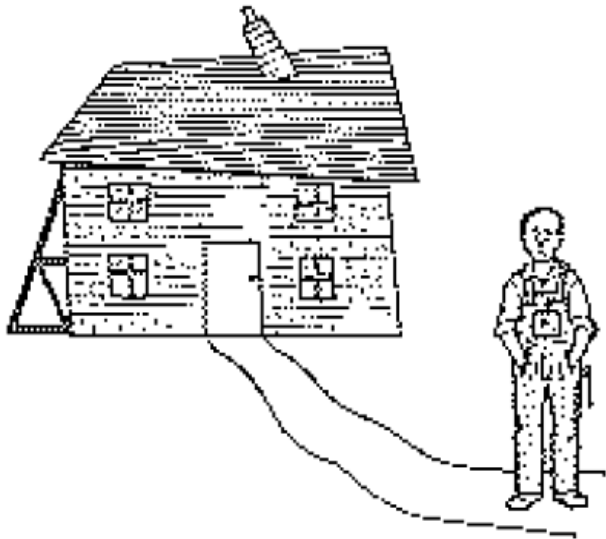


GIS Design and Implementation

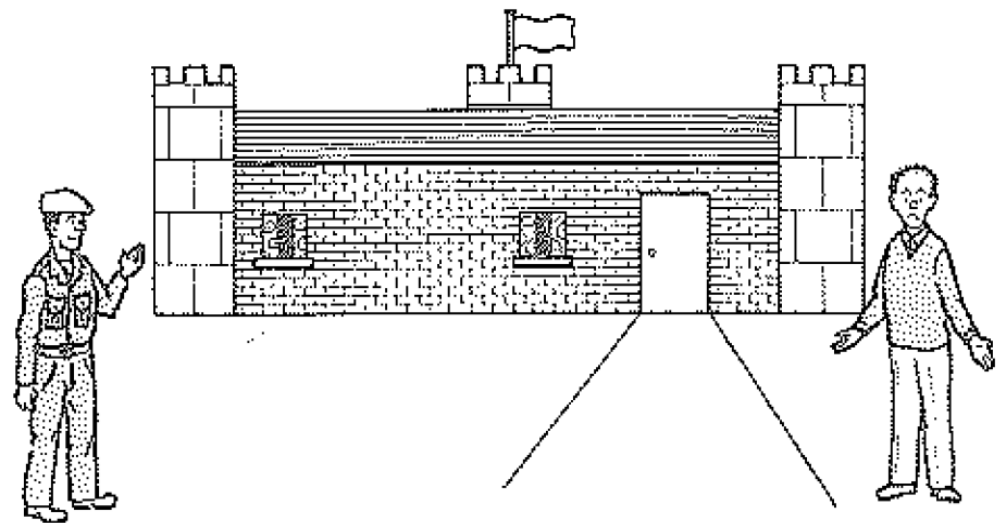


What is System Analysis and Design

- Assume you have bought a plot of land in order to build yourself a house.



Option 1: Start constructing it straightaway.



Option 2: Go directly to a builder.

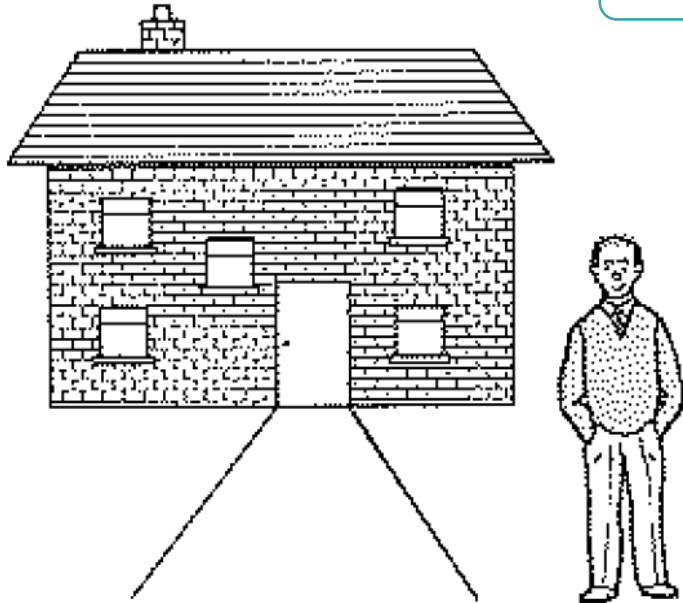


GIS Design and Implementation



What is System Analysis and Design

Option 3: Employ an architect to design it first.



Tell the specific requirements.

Point out the regulations and constraints.

Develop several design options for the owner to choose.

Translate the owner's requirements into designs which can be read by the builders.



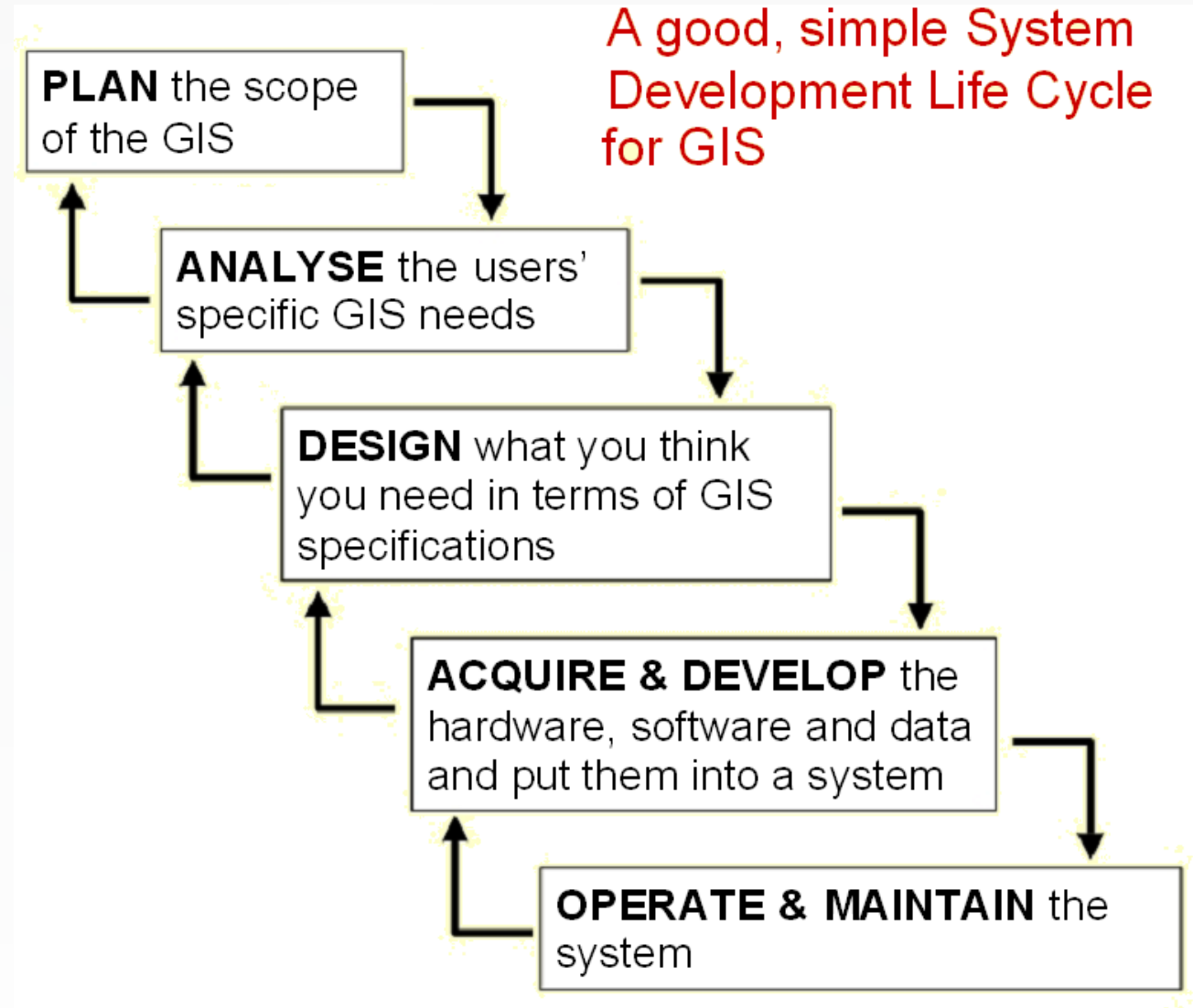
GIS Design and Implementation

- **System Analysis and Design**

- **To translate the users' requirements into specific designs of information system.**
- **Methods that structure a project into small, well defined activities**
 - **Used by the HKSAR Govt. as the standard in System Development (including GIS)**
 - **http://www.ogcio.gov.hk/en/infrastructure/methodology/software_life_cycle/systems_analysis_and_design.htm**



GIS Design and Implementation

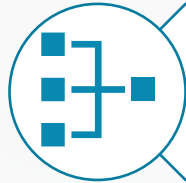




GIS Design and Implementation



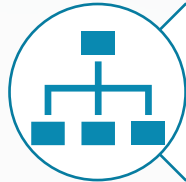
What is Planning?



Process of aligning GIS implementation and operations with the organization's business priorities



Creating the long-term big picture—the target for GIS implementation



Defining the role of the GIS within the entire organization and its information technology architecture

Find the problems within the current organization and make a plan to solve the problems using GIS and related technologies.



Scope of GIS Projects

Single-purpose GIS project

- Initiated to serve one-time objectives

Departmental GIS project

- A department is responsible for implementing GIS functionalities

Enterprise GIS project

- Many functions are shared between different departments, thus implementing GIS projects across multiple departments make sense.
- Will benefit from system integration.
- Increasing strategic roles of GIS in an organization.

Societal GIS project

- Cost sharing by the society as a whole



User Requirement Analysis

- Determine how the GIS application system is designed and developed
- Focus on: what information is currently being used, who is using it, how the source is being collected, stored, and maintained, and is there any new demand.
- Necessary information can be obtained through interviews, documentations, reviews, and workshops



System Design

Functional design

- Modules involved
 - Relationship between the modules

Database design

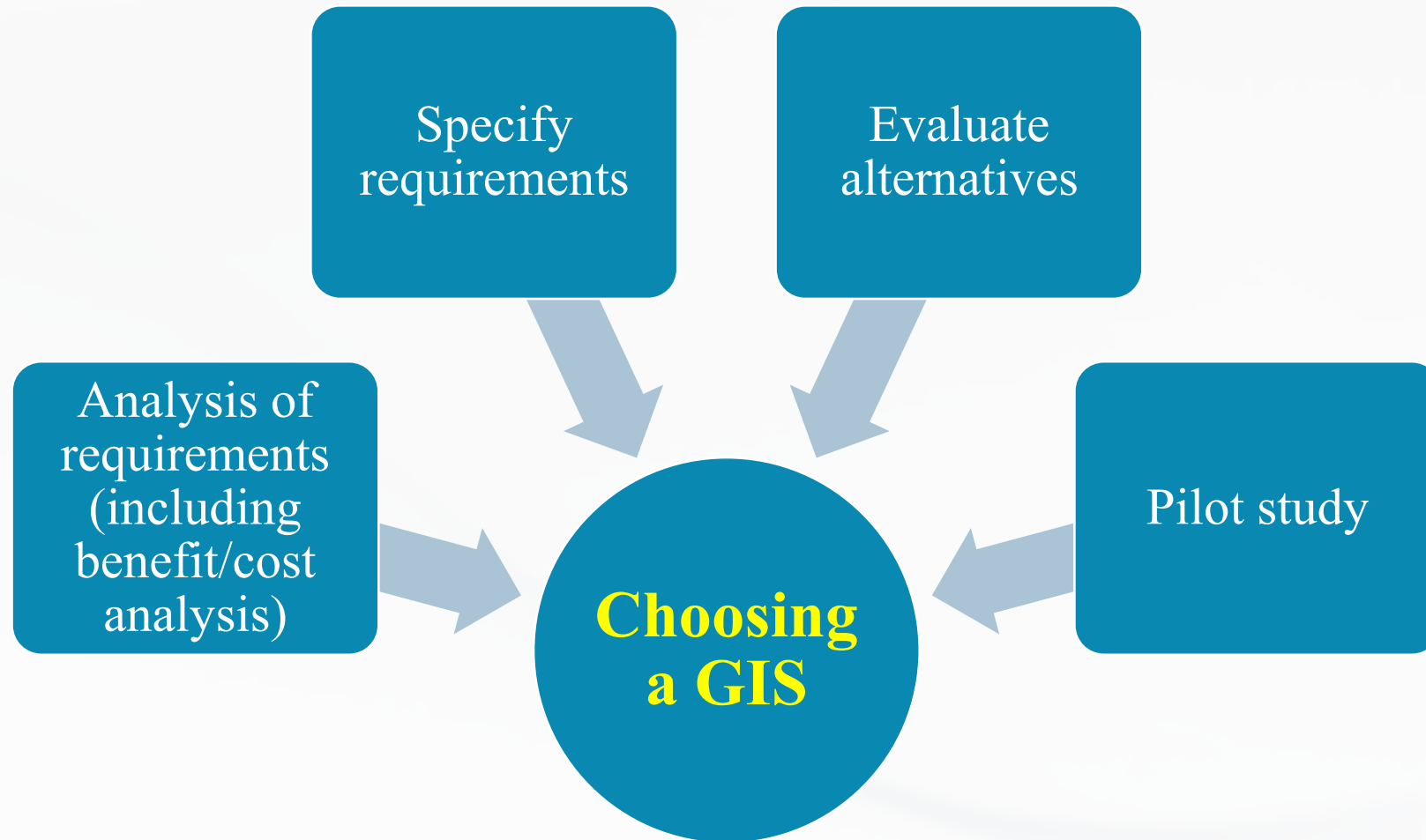
- Conceptual design
- Logical design
- Physical design

User Interface Design

- Usability



System Acquisition and Implementation





GIS Implementation Issues

**Ensure a robust
system**

**Ensure database
quality and security**

**Accommodate GIS
within organization**

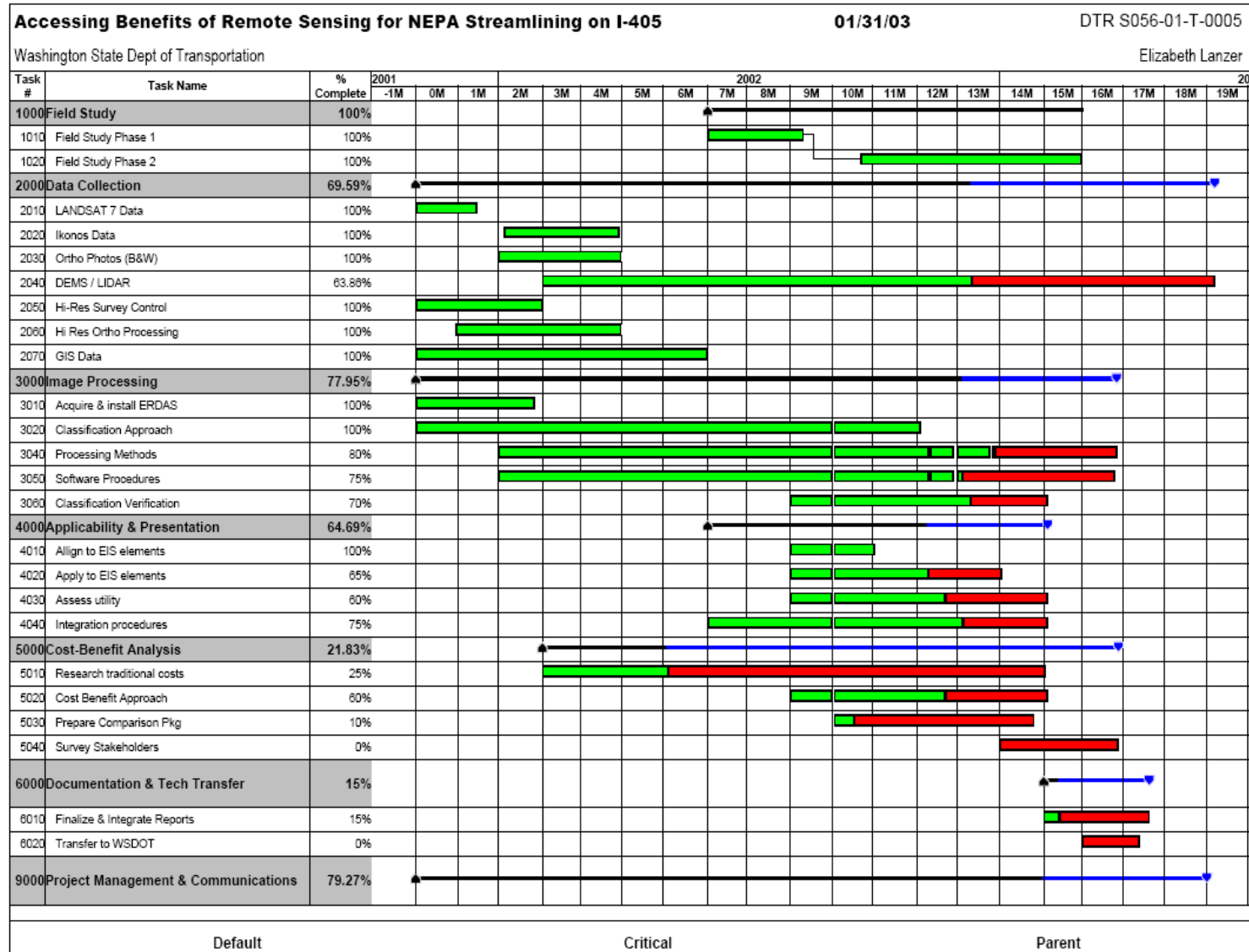
**Avoid unreasonable
timeframes**

**Secure ongoing
funding**

Prevent meltdown



GANTT Chart



Courtesy:
Washington State
Department of
Transportation

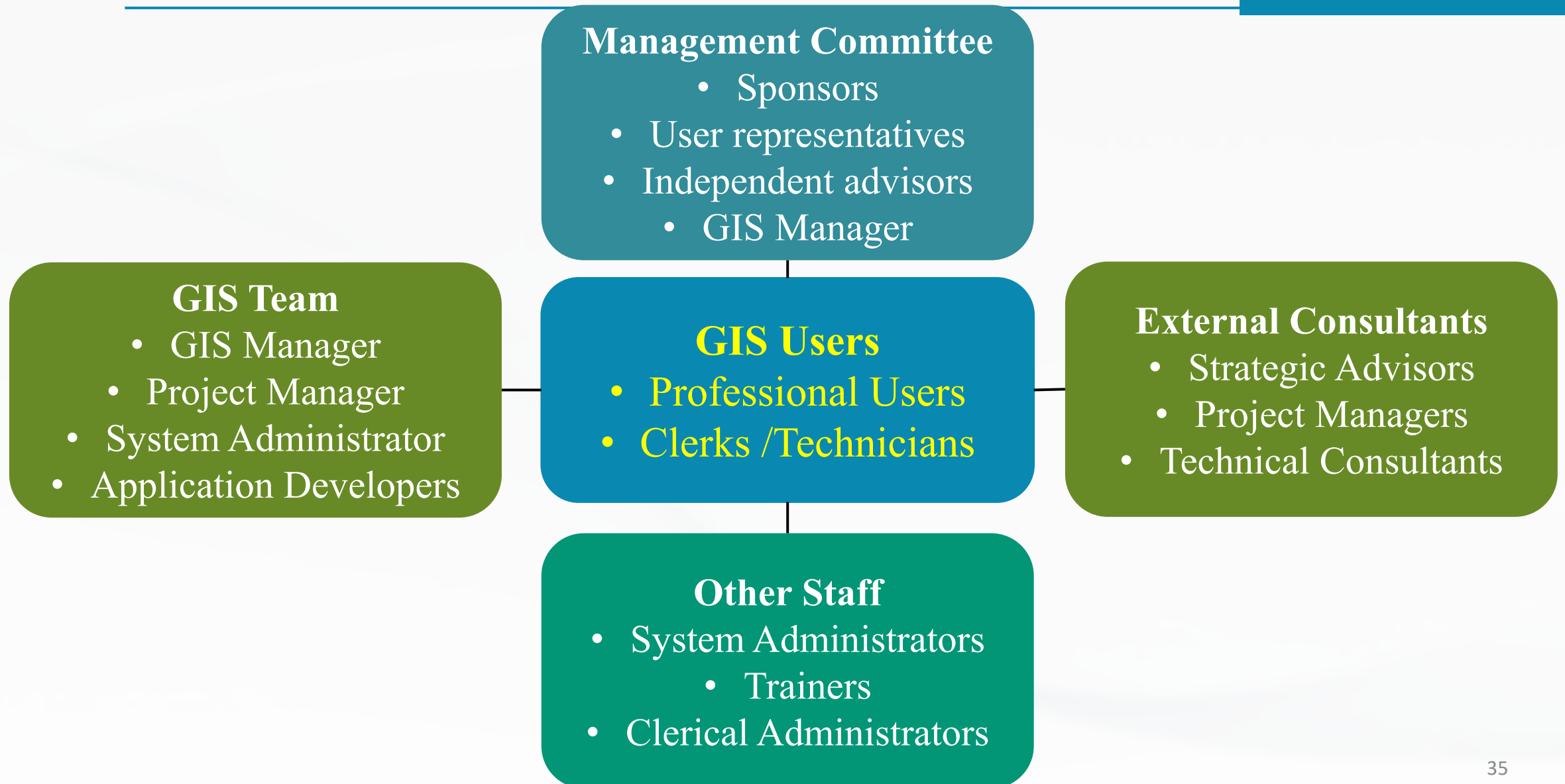


Operations of a GIS

- **Customer support**
 - All users are customers
 - Create customer support facility
- **Operations support**
 - Administration, backup, system support
 - Helpdesk
- **Data management support**
 - Database Administrator
- **On-going application customization**
- **Use well-proven project management tools**



GIS Staff Roles





Golden Rules of Project Management

Projects must be completed on **time**, within **budget**, and according to quality **standards**

You will be responsible for the work of others. Make sure they are competent

Uncertainty of many kinds exists: you have to live with it but agree how much is acceptable

Have fun doing it and celebrate success!



Reasons GIS Fail

- Lack of executive-level commitment
- Inadequate oversight of key participants
- Inexperienced managers
- Unsupportive organizational structure
- Political pressures, e.g. in times of fast change
- Inability to demonstrate benefits
- Unrealistic deadlines
- Poor planning
- Lack of core funding



Summary

- GIS design and implementation is much the same as other IT
 - Mixture of technical and people issues; People ones are often trickier
- Involves motivating, organizing, monitoring: some of it is in common sense

Most important thing is to create, nurture and grow a good team